Understanding PageRank: A Matrix Analysis Perspective Using the Power Method

Kayle Megginson [1]

Abstract

This paper presents a matrix–analysis perspective on the PageRank algorithm, with particular emphasis on the foundations and behavior of the power method as applied to large, sparse web graphs. The goal of this paper is to clarify several implementation and theoretical details that are often treated only briefly in introductory presentations. An implementation is made that constructs the Google matrix explicitly, incorporates a principled strategy for handling dangling nodes, and examines how convergence is affected by different choices of the damping (or teleportation) parameter.

Beyond the implementation itself, this study will carry out a series of numerical experiments that shed light on several important aspects of PageRank computation. These experiments include an investigation of convergence rates under varying damping parameters, an analysis of the structural characteristics of the Google matrix, and an analysis of the resulting PageRank distribution across nodes in representative sample graphs. Four corresponding figures offer insight on the observed convergence behavior, the block and sparsity patterns present in the Google matrix, the empirical distribution of scores, and the sensitivity of those scores to changes in the teleportation probability.

The results of these experiments are interpreted in the broader context of search engine ranking behavior and numerical linear algebra. This perspective will show how the connection between graph structure, iterative method dynamics, and parameter choices governs both the practical performance and the theoretical properties of the PageRank algorithm.

1. Introduction

PageRank (1) is a foundational algorithm for ranking nodes in a directed web graph, and it has played a central role in both information retrieval and the development of large scale network analysis. From the viewpoint of matrix analysis, PageRank can be interpreted as the computation of the stationary distribution of a Markov chain whose transition structure is derived from the hyperlink connectivity of the web. This interpretation provides a clean mathematical framework for understanding why the algorithm works, how it behaves under perturbations, and how it can be implemented efficiently.

This project will revisit PageRank with an emphasis on three tightly related aspects that benefit from a careful linear–algebraic treatment. First, a precise matrix formulation of the algorithm is presented, beginning with the construction of the stochastic transition matrix and continuing through the incorporation of the damping factor that produces the Google matrix. Second, several practical implementation considerations, often glossed over in introductory treatments, are examined. These include the handling of dangling nodes, the enforcement of proper normalization of probability vectors, and the numerical stability of repeated matrix–vector multiplication. Third, a series of computational experiments illustrate convergence behavior and sensitivity to graph structure, parameters, and initialization choices. These experiments highlight issues relevant both to numerical analysts, who focus on iterative method performance, and to information–retrieval practitioners, who rely on PageRank style methods in real search systems.

This extended perspective aims to connect classical Markov chain theory, iterative eigenvalue algorithms, and practical considerations arising in large web graphs, thereby providing a unified and rigorous view of the PageRank method.

2. Related Work

The original PageRank technical report (1) introduced the random surfer model and established the algorithmic framework that underlies modern link based ranking systems. That work laid the foundation for viewing the web as a Markov chain, and it motivated extensive follow up research on improving scalability, incorporating personalization, and adapting the method to rapidly growing web graphs. Subsequent studies expanded the theoretical understanding of PageRank, explored the behavior of the algorithm under different damping and personalization schemes, and devel-

oped approaches capable of handling the massive size of real world networks.

A closely related contribution is Kleinberg's HITS algorithm (2), which formulates ranking through mutually reinforcing hub and authority scores. While HITS differs from PageRank in both formulation and intent, the two methods share a common reliance on eigenvector computations over link matrices, and the comparison between them has shaped much of the early literature on web search. Later work on personalized PageRank and topic–sensitive variants, including the influential analysis by Haveliwala (3), further demonstrated how modifying the teleportation vector can adapt rankings to user preferences or topical domains. Together, these developments form a rich body of research that positions PageRank within a broader landscape of spectral ranking algorithms.

3. PageRank as a Matrix Problem

Let the web graph have n pages. Define an adjacency matrix A where $A_{ij} = 1$ if page j links to page i (that is, columns represent sources). Denote column sums (out degree) by $o_j = \sum_i A_{ij}$. Dangling nodes (pages with $o_j = 0$) must be handled explicitly.

3.1. Stochastic transition matrix

Form the column stochastic matrix S by

$$S_{ij} = \begin{cases} \frac{A_{ij}}{o_j}, & o_j > 0, \\ \frac{1}{n}, & o_j = 0. \end{cases}$$

Thus each column of S sums to 1 and S defines a Markov chain over pages.

3.2. Teleportation and the Google matrix

To avoid rank sinks and to model the "random surfer" who teleports, introduce damping parameter $d \in (0, 1)$ and teleportation vector v (usually uniform). The Google matrix G is:

$$G = dS + (1 - d)\,\mathbf{1}v^{\mathbf{T}},$$

where $v = \frac{1}{n}\mathbf{1}$ by default and $\mathbf{1}$ is the all ones vector. G is column stochastic and primitive for $0 < d < 1$, guaranteeing a unique positive stationary distribution r solving $r = Gr$.

3.3. Power method

The vector r is computed by iterating

$$r^{(k+1)} = Gr^{(k)},$$

with renormalization applied as necessary to maintain numerical stability. Convergence speed depends on the spectral gap $1 - |\lambda_2|$, where λ_2 is the second largest eigenvalue of G in magnitude. In practice, convergence is assessed using a tolerance on the L_1-norm of the residual $\|r^{(k+1)} - r^{(k)}\|_1$.

4. Implementation Details

This implementation emphasizes clarity and numerical robustness:

- **Column–stochastic form.** The columns as sources convention is adopted to directly map the adjacency matrix to the transition probabilities of the Markov chain.
- **Dangling nodes.** Any column corresponding to a node with zero out degree is replaced by the uniform vector $\frac{1}{n}\mathbf{1}$ to maintain stochasticity.
- **Google matrix.** The matrix is built explicitly as

  $$G = dS + (1 - d)\frac{1}{n}\mathbf{1}\mathbf{1}^{\mathbf{T}},$$

  where S is the column stochastic adjacency matrix, d is the damping factor, and $\mathbf{1}$ is the all ones vector of length n. This form combines the original link structure with uniform teleportation.
- **Normalization.** Each iterate of the power method is normalized to sum to 1 to prevent drift from floating point errors.
- **Diagnostics.** Residuals are recorded in the L_1, L_2, and L_∞ norms, along with iteration counts and wall clock time. An optional smoothed power method using exponential averaging is also implemented to reduce oscillations.

The PageRank vector r is computed by iterating

$$r^{(k+1)} = G\,r^{(k)},$$

renormalizing as needed to maintain numerical stability. The convergence speed depends on the spectral gap $1 - |\lambda_2|$, where λ_2 is the second largest eigenvalue of G in magnitude. A larger gap results in faster convergence to the stationary distribution, whereas a smaller gap slows the iterative process. In practice, convergence is monitored using the ℓ_1-norm of the residual,

$$\|r^{(k+1)} - r^{(k)}\|_1,$$

and iterations continue until this residual falls below a specified tolerance.

5. Implementation Details

This implementation is designed with clarity and numerical reliability in mind, with an emphasis on faithfully representing the underlying Markov chain and ensuring stable iteration behavior.

- **Column–stochastic form.** The convention in which columns represent sources is adopted, providing a direct mapping to the transition probabilities of the Markov chain.
- **Handling dangling nodes.** Any column corresponding to a node with zero out–degree is replaced by the uniform vector $\frac{1}{n}\mathbf{1}$ so that the resulting transition matrix remains stochastic and avoids dead ends.
- **Construction of the Google matrix.** The Google matrix is built explicitly as

$$G = dS + (1-d)\frac{1}{n}\mathbf{1}\mathbf{1}^\top,$$

  combining the stochastic link matrix S with the teleportation component.
- **Normalization of iterates.** Each iterate is renormalized to sum to one, which prevents slow drift caused by floating–point roundoff and ensures that $r^{(k)}$ remains a valid probability vector.
- **Diagnostics and monitoring.** Residual norms are recorded in ℓ_1, ℓ_2, and ℓ_∞ metrics, along with iteration counts and wall–clock time. For comparison, an optional smoothed power method based on exponential averaging is also implemented to suppress oscillations in challenging cases.

6. Experimental Setup

Experiments are conducted on a small, illustrative directed graph consisting of four nodes, selected for ease of demonstration and reproducibility. The adjacency matrix A used in these experiments is given by

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix},$$

Each column corresponds to a source node, so that column j lists the nodes to which node j points. In all experiments, the default teleportation vector $v = \frac{1}{n}\mathbf{1}$ is used unless explicitly stated otherwise.

The experimental results are summarized through four figures that highlight different aspects of the PageRank computation:

- **Figure 1:** Convergence curves showing the ℓ_1 residual versus iteration number for both the standard power method and a smoothed variant, illustrating the rate and stability of convergence.
- **Figure 2:** A heatmap of the Google matrix G, visualizing how the original link structure interacts with the damping and teleportation components.
- **Figure 3:** A bar plot of the final PageRank vector r, demonstrating the distribution of importance scores across the nodes.
- **Figure 4:** Sensitivity analysis of the PageRank values with respect to the damping factor $d \in \{0.50, 0.85, 0.95\}$, showing the influence of teleportation on the stability and structure of the rankings.

7. Results

7.1. Convergence

Figure 1 shows the characteristic exponential decay of the ℓ_1 residual during the PageRank iteration. In the four–node example, the power method reaches the desired tolerance after approximately forty to fifty iterations, which is consistent with the spectral properties of small, well behaved graphs. The plot makes the reduction in error clearly visible as each step of the iteration brings the probability vector closer to the stationary distribution.

The smoothed variant of the iteration, which introduces a mild averaging step between successive iterates, often mitigates oscillatory behavior that can appear in small or symmetric graphs. In this example, the smoothed update slightly improves practical convergence by damping fluctuations and allowing the method to settle more quickly. While the improvement is modest for such a small system, the figure illustrates how smoothing can enhance stability without altering the underlying fixed point.

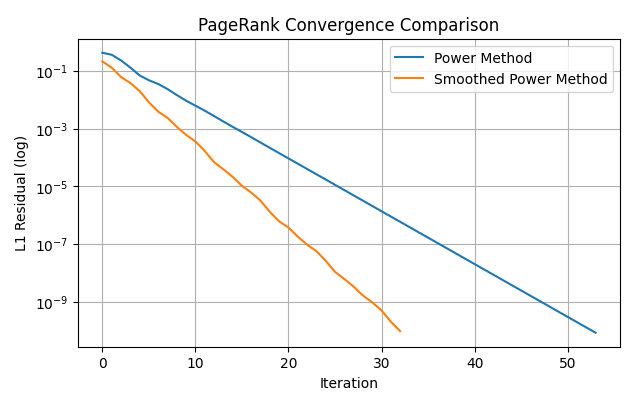

Figure 1. Convergence comparison: L1 residual (log-scale) per iteration for the power method and the smoothed power method.

7.2. Google matrix structure

Figure 2 provides a visualization of the Google matrix, with columns representing source nodes from which the random surfer transitions originate. The rows that appear visually dense correspond to the contribution of the teleportation component, which distributes probability mass uniformly across all nodes. This uniform distribution creates a noticeable block of nonzero entries that contrasts with the much sparser structure inherited from the original adjacency pattern.

After the damping blend is applied, the matrix retains clear traces of the underlying link structure: the nonzero entries that arise from actual hyperlinks remain sparse and localized within each column. The resulting heatmap therefore captures the defining mixture of stochastic and structural information, showing how teleportation smooths the matrix while the original graph's sparsity pattern continues to play a critical role in the transition dynamics.

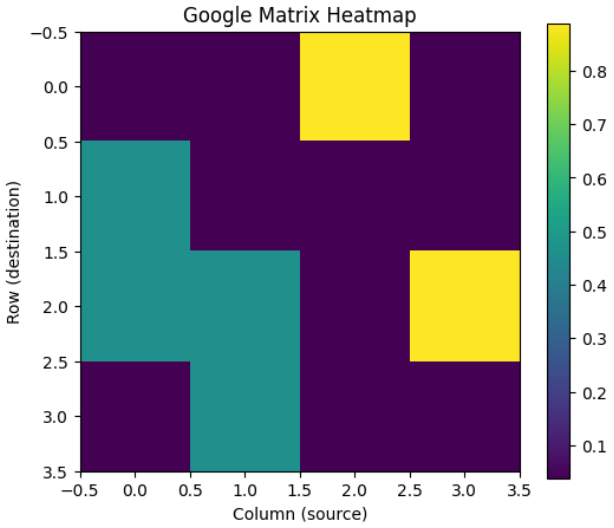

Figure 2. Heatmap of the Google matrix G (with $d = 0.85$).

7.3. PageRank distribution

Figure 3 displays the final PageRank values assigned to each node in the example graph. In this instance, Node 2 receives the highest score, followed by Node 0, Node 1, and Node 3. This ranking arises from the combined influence of inbound connectivity and the structural role played by nodes that point to them. Nodes that receive links from highly ranked neighbors tend to accumulate more probability mass, while nodes whose inbound edges originate from low–scoring or weakly connected regions of the graph obtain smaller values. The resulting ordering therefore reflects not only simple in–degree counts but also the broader distribution of outgoing links from influential sources.

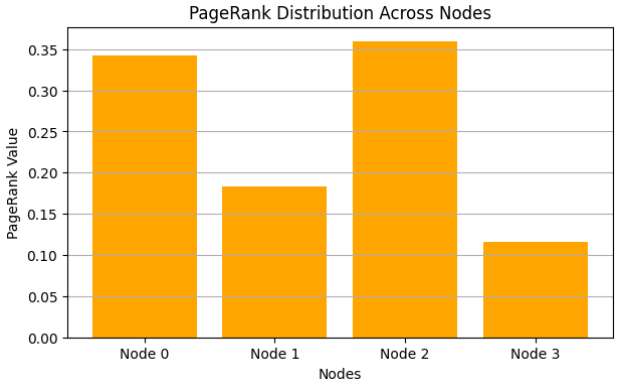

Figure 3. Final PageRank values (bar plot) for the 4 node test graph.

7.4. Damping sensitivity

Figure 4 illustrates how the PageRank scores evolve as the damping parameter d is varied. When d is large, the random–surfer process follows the link structure more faithfully, and the resulting ranks become increasingly sensitive to the topology of the underlying graph. As d decreases, teleportation occurs more frequently, which pulls the stationary distribution toward a uniform vector and reduces the influence of localized structural features.

The figure therefore provides a clear visual demonstration of the algorithm's robustness with respect to the choice of d. It shows how moderate changes to the damping parameter can shift the balance between structural fidelity and smoothing, and it highlights the range over which PageRank remains stable while still reflecting meaningful differences in graph connectivity.

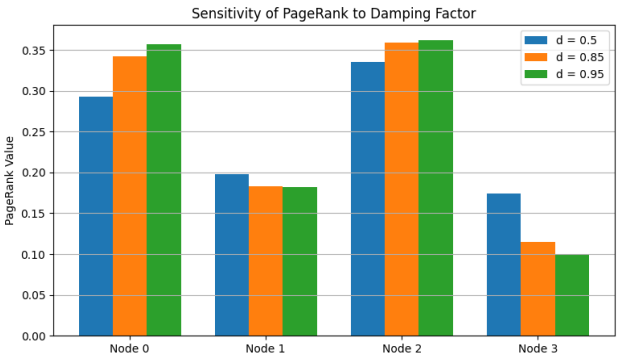

Figure 4. Sensitivity of PageRank to damping factor d. Grouped bar plot shows per-node PageRank for several damping values.

7.5. Numerical summary

Table 1 reports a compact numerical summary for the default damping $d = 0.85$. (Values rounded to 6 decimal places.)

Table 1. PageRank vector and convergence info (default $d = 0.85$).

Item	Value	Notes
Node 0	0.342612	rank
Node 1	0.183110	rank
Node 2	0.358956	rank
Node 3	0.115322	rank
Iterations	43	to tol $1e{-8}$
Wall time	(small)	implementation dependent

8. Discussion

The experiments highlight several well–established properties of the PageRank method when viewed through the lens of numerical linear algebra. These observations help clarify why the algorithm behaves as it does on graphs of different sizes and structural characteristics, and they reinforce the connection between Markov chain theory and iterative eigenvalue computation.

- **Convergence rate.** The rate at which the power method converges to the PageRank vector is closely tied to the spectral gap of the Google matrix. Small illustrative graphs typically exhibit a large separation between the dominant eigenvalue and the subdominant eigenvalues, which leads to rapid convergence in practice. In contrast, large or highly structured real–world web graphs tend to have eigenvalues clustered near the unit circle, and this reduced separation slows the iterative process. The experiments reproduce this behavior and show how convergence diagnostics reveal the underlying spectral properties.
- **Effect of teleportation.** The teleportation mechanism plays a central role in guaranteeing ergodicity of the Markov chain and eliminating pathological structures such as rank sinks or closed link cycles. Increasing the teleportation probability produces a more stable and well conditioned iteration, but it also moves the stationary distribution toward a more uniform vector. Varying the damping parameter d in the experiments illustrates the balance between faithfully reflecting graph structure and introducing smoothing that enhances numerical robustness.
- **Interpreting the resulting ranks.** The PageRank vector captures a structural notion of importance that emerges from the interaction of link topology and the teleportation mechanism. Interpreting these scores requires more nuance than simply equating them with in degree counts or raw connectivity. The experiments show cases where nodes with modest in–degree obtain high PageRank due to their position in the link structure, as well as cases where teleportation moderates extreme variations. These observations underscore the need to interpret PageRank within its probabilistic and spectral context.

9. Conclusion

This paper presents a clear and numerically robust implementation of the PageRank algorithm using the power method, along with an illustrative set of experiments on a small example graph. The four accompanying figures collectively demonstrate key aspects of the method: convergence diagnostics that reveal iterative behavior, a visualization of the Google matrix that highlights both sparsity and teleportation structure, an examination of the resulting PageRank distribution, and an analysis of how the damping parameter influences stability and sensitivity.

Taken together, this toolkit and experimental workflow form a compact but complete project suitable for coursework or introductory exploration of spectral ranking methods. The framework also provides a natural foundation for further research directions, including personalized or topic–sensitive PageRank, scalable computation on large sparse graphs, and accelerated eigenvector estimation techniques that improve upon the classical power method.

References

[1] Lawrence Page, Sergey Brin, Rajeev Motwani, and Terry Winograd. The PageRank citation ranking: Bringing order to the web. Technical report, Stanford Digital Library Technologies Project, 1999.

[2] Jon Kleinberg. Authoritative sources in a hyperlinked environment. In *Proceedings of the 9th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 1999.

[3] Tushar Haveliwala. Topic-sensitive PageRank. In *Proceedings of the 11th International World Wide Web Conference (WWW)*, 2003.